

Causal Emergence in U.S. Equity Industry Portfolios: Dynamic Organization and Effective Dimensionality During Systemic Stress

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Declaration of Competing Interests

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and Code Availability

All datasets used in this study (Kenneth French 30 and 49 Industry Portfolios) are publicly available from the Kenneth French Data Library. Complete replication code, numerical optimization routines, surrogate null generators, and reproduction pipelines are maintained in the project repository at <https://github.com/ShiroMorph/causal-emergence-financial-risk> and will be released under an open-access DOI upon publication.

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Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author used OpenAI GPT in order to edit language, improve readability, and adapt the writing of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.